

Errors and Analysis

Proposed Data Improvements for Model Iteration

Based on the initial error analysis of the YOLOv11 crack detection model, the following three data-driven strategies are proposed to enhance accuracy and reliability in the subsequent version.

1. Hard Negative Mining for False Positive Reduction

- **The Problem:** The current model frequently confuses non-structural lines (such as overhead power lines, shadows from trees, and expansion joints) with actual cracks.
- **The Improvement:** We will incorporate "Background Images" (images containing no cracks but featuring "distractor" objects like wires, shadows, and rebar) into the training set.
- **Goal:** By providing the model with examples of what a crack is not, we force it to learn more granular features, effectively reducing the False Positive rate.

2. Environmental & Lighting Augmentation (Low-Light/Wet Surfaces)

- **The Problem:** The model shows decreased sensitivity (False Negatives) in images taken during overcast conditions or on wet concrete surfaces where the contrast between the crack and the substrate is low.
- **The Improvement:** We will implement a targeted augmentation pipeline that specifically applies "Random Brightness," "Contrast Shifting," and "Blur" to the existing dataset. Additionally, we will seek out 200-300 new samples specifically showing facades during rain or at dusk.
- **Goal:** This will improve the model's robustness, ensuring it can reliably detect structural defects regardless of weather or time-of-day constraints.

3. Multi-Scale Tiling for Hairline Crack Detection

- **The problem:** The standard 640x640 input resolution often "washes out" thin hairline cracks, causing the model to miss early-stage structural issues (False Negatives).
- **The Improvement:** We will implement Slicing Aided Hyper Inference (SAHI) or simple Image Tiling. Instead of resizing a large facade photo to a small square, we will divide the high-resolution original image into smaller, overlapping tiles (e.g., 320x320) for training and inference.
- **Goal:** This preserves the high-frequency details of the crack, allowing the model to detect micro-cracks that are currently invisible at lower resolutions.